

Data Vortex
Aaruush '26
Round-1 (Phase-2)
Questioner



EASY LEVEL

E1 — Platform Popularity

Challenge:

Determine which social media platform has the highest number of posts. Ignore posts where the platform is missing. Return the platform name and number of posts.

E2 — Most Engaged Posts

Challenge:

Find the top 10 posts based on total engagement. Total engagement is defined as likes + shares + comments. Ignore posts where likes are missing.

E3 — Average Engagement by Platform

Challenge: Calculate the average likes, shares, and comments for each platform. Which platform generates the highest average total engagement?

E4 — Highly Shared but Poorly Liked

Challenge:

Identify posts that received more than 1,500 shares but fewer than 500 likes. Return the post ID, platform, likes, shares, and comments.

E5 — Users With Large Audiences

Challenge: Find all users with more than 40,000 followers. Display their user ID, location, language, and follower count.

MEDIUM LEVEL

M1 — Which Locations Generate the Most Engagement?

Challenge:

Using both datasets, calculate the total engagement generated by users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement.

M2 — Do High Follower Users Get More Engagement?

Challenge: Divide users into two groups:

- High follower users: $\geq 25,000$ followers
- Low follower users: $< 25,000$ followers

Compare their average engagement per post.

M3 — Most Active Users

Challenge:

Find the top 10 users who have created the highest number of posts. Display their follower count, location, post count, and total engagement.

M4 — Platform Behaviour by High Follower Users

Challenge:

Among users with at least 30,000 followers, determine which platform gives them the highest average engagement per post.

M5 — Detect Suspicious Engagement

Challenge:

Find posts where the number of shares is greater than the number of likes and comments combined. Such posts may represent unusual sharing behaviour. Return the top 20.

HARD LEVEL

H1 — Find Users With Abnormally High Engagement

Challenge:

Calculate the average total engagement per user. Identify users whose average engagement per post is more than twice the overall average engagement per post.

Return the user ID, location, follower count, post count, and average engagement.

H2 — Rank Users Within Their Location

Challenge:

For every location, rank users based on their total engagement. Return only the top 3 users from each location.

H3 — Platform Performance Compared With Its Own Average

Challenge:

For every platform, identify posts whose engagement is significantly higher than the average engagement of that platform.

A post is considered exceptional if its engagement is at least $2\times$ the average engagement of its platform.

H4 — Follower to Engagement Anomaly

Challenge:

Find users who have fewer than 5,000 followers but whose total post engagement places them among the top 10% of all users.

These users may represent unusually high performing or suspicious accounts.

This requires multiple levels of analysis.

H5 — Identify Data Anomalies

This one specifically exploits your corrupted dataset.

Challenge:

Identify potentially corrupted posts where one or more of the following conditions are true:

1. Likes are negative.
2. Platform is missing.
3. Text content is missing.
4. Text contains HTML entities/tags such as `&`, `<div>`, or `<br>`.

Return the post ID and identify the type of anomaly.

H6 — Find the Most Suspicious High Impact Users

Challenge:

Identify users who satisfy all three conditions:

- Have fewer than 10,000 followers.
- Their average post engagement is above the overall average.
- At least one of their posts has more shares than likes.

Rank these users by total engagement and display their location, follower count, number of posts, average engagement, and total engagement.